VISHWAS KOTHARI

M.S. COMPUTER SCIENCE | ML EVALUATION | DATA SYSTEMS | RESPONSIBLE AI

+1 (720) 388-4856 | vishwasvkothari@gmail.com | linkedin.com/in/vishwas-kothari | github.com/vishwas182002 | Portfolio

PROFILE

M.S. Computer Science student (May 2027) who builds machine learning and data systems end to end, then stress-tests them before they ship. Research background at ISRO in numerical optimization and satellite data pipelines; recent work centers on model evaluation, failure analysis, and explainability — locating where systems break, not where they score well.

EDUCATION

UNIVERSITY OF COLORADO BOULDER — **Boulder, CO | Aug 2025 – May 2027**

Professional Master's in Computer Science | GPA: 3.74/4.0

Coursework: Neural Networks & Deep Learning, Theory of Machine Learning, Design & Analysis of Algorithms, Database Systems, Linear Programming

VELLORE INSTITUTE OF TECHNOLOGY — **Vellore, India | Aug 2021 – Jun 2025**

B.Tech in Information Technology | GPA: 8.49/10.0

EXPERIENCE

SPACE APPLICATIONS CENTRE, INDIAN SPACE RESEARCH ORGANISATION (ISRO) — **Ahmedabad, India**

Research Intern, Biological Oceanography Division | Jan 2025 – Jul 2025

- **Compared** local and global optimization by implementing Levenberg–Marquardt and a custom Particle Swarm Optimization on NASA NOMAD; LM reached R^2 = 0.987 on reflectance reconstruction and log-space R^2 = 0.879 on chlorophyll-a validation (n = 105).
- **Stabilized** convergence across constituents spanning three orders of magnitude by introducing log-space parameter transformation and bounded search constraints, resolving instability and high-concentration bias in the baseline method.
- **Automated** OCM/MODIS Aqua cross-sensor validation with a Python pipeline (NumPy, Pandas, xarray, SNAP) for EPSG:4326 reprojection, resampling, date-based pairing, pixel-level colocation, and NetCDF/CSV outputs supporting OCM-3 mission validation.
- **Standardized** research handoff through four agreement metrics (RMSE, MAE, R^2, bias), six diagnostic visualization types, and a 44-page technical report presented to cross-division staff.

SELECTED PROJECTS

FINAGENT-EVAL | GitHub

Python, PyTorch, Hugging Face, Qwen2.5-VL, Streamlit, GitHub Actions, pytest | 2026

- **Benchmarked** five vision-language and document-AI models on 397 financial-document questions, raising ANLS from 0.147 to 0.637 (4.3x) with a 4-bit quantized Qwen2.5-VL-7B baseline on a single T4 GPU.
- **Exposed** reliability limits hidden by aggregate accuracy: only 4 of 65 numerical-reasoning questions correct despite 59% overall exact match, and 44% answer consistency under paraphrase.
- **Shipped** 12 CI-gated tests, 1,000-sample bootstrap confidence intervals, a seven-label failure taxonomy, and a deployed seven-tab dashboard for independent verification.

FINANCIAL DOCUMENT VQA | GitHub

Python, PyTorch, LayoutLMv3, Donut, Pix2Struct, LoRA/PEFT, Tesseract OCR | 2026

- **Curated** a 397-question SEC-filing benchmark from 79 images across 10 companies and measured a 63–83% drop from generic-benchmark scores, surfacing a ranking reversal in which OCR-based LayoutLMv3 beat OCR-free models on dense tables.
- **Raised** held-out ANLS from 0.097 to 0.134 via domain-specific LoRA on 307 examples evaluated on three unseen companies, and documented 39% negative transfer from generic DocVQA fine-tuning.

XAI CREDIT LENS | GitHub

Python, Scikit-learn, XGBoost, LightGBM, Optuna, SHAP, LIME, DiCE, Streamlit | 2026

- **Achieved** 0.782 test ROC-AUC on 30,000 credit records by tuning XGBoost, LightGBM, and Random Forest across 100 Optuna trials with stratified five-fold cross-validation.
- **Integrated** SHAP, LIME, and DiCE to surface decision drivers and generate counterfactual recourse restricted to applicant-actionable features, excluding immutable attributes by design.
- **Audited** fairness across four protected attributes (disparate-impact ratios 0.960–1.029) and built governance modules mapped to ECOA, EU AI Act, and Fed SR 11-7.

IMPLICIT BIAS OF ADAM VS. SGD | GitHub

Python, PyTorch, CVXPY, NumPy, controlled robustness testing | 2026

- **Verified** optimizer implicit bias against exact convex max-margin solutions across five seeds, reaching cosine similarities of 0.990 for gradient descent (L2) and 0.988 for Adam (L-infinity).
- **Measured** a 30.2-point out-of-distribution accuracy drop for SGD versus 13.6 for Adam under a spurious correlation flipped from 99% to 1%, despite near-identical in-distribution accuracy.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Java, C/C++, JavaScript, NumPy, Pandas, SciPy, xarray, NetCDF, Matplotlib, Plotly
ML & Deep Learning: PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost, LightGBM, LoRA/PEFT, Optuna, CVXPY
Evaluation & Interpretability: Bootstrap confidence intervals, failure taxonomies, ANLS/exact-match grading, SHAP, LIME, DiCE, fairness auditing
Data Systems (course projects): Apache Spark, PySpark, Spark SQL, Redis, PostgreSQL, Node.js
Engineering & Applications: Git, GitHub Actions (CI), pytest, Linux, Jupyter, Streamlit, Gradio, AWS, LaTeX

PUBLICATION, CERTIFICATION & SERVICE

Publication: Kothari, V. and Krishwanth, B. (2025). *Empowering Survivors: Ethical AI for Countering Violence Against Women.* Springer.
Certification: AWS Certified Cloud Practitioner — 970/1000 (2024).
Peer Reviewer: *Public Health*; *Social Sciences & Humanities Open* (Elsevier), 2026–Present.